

[← BACK TO INDEX](#)

01 Document Q&A with citations

• BEGINNER

Python

LangChain

Chroma

~3h

6 steps

Start Building →

Save

DEPTH

Beginner

Intermediate

Advanced

Local files, a single PDF, and a prebuilt retriever – you wire the pieces together and see it answer with citations.

What It Does

Build It

Step 0 of 6

0%

01

Set up the environment

Install Python deps and load your PDF into the project.



02

Load and split the document

Read the PDF and break it into overlapping chunks.



03 Embed and index the chunks

Turn each chunk into a vector and store it in Chroma.

Each chunk is embedded into a vector and written to a local Chroma collection. This index is what makes retrieval fast — build it once, query it many times.

```
from langchain_community.vectorstores import Chroma
from langchain_openai import OpenAIEmbeddings

embeddings = OpenAIEmbeddings()
db = Chroma.from_documents(
    documents=chunks,
    embedding=embeddings,
    persist_directory="./chroma",
)
db.persist()
print(f"Indexed {len(chunks)} chunks")
```

04 Build the retriever

Fetch the top-k most relevant chunks for a query.

05 Wire up the Q&A chain

Pass retrieved context to the model and return citations.

06 Ask questions & verify sources

Run a query and click through to the cited passages.

TIME REMAINING

~3h

of ~3h estimated

 **Stuck? Ask Claude**

Paste an error or describe where you're blocked — get an answer scoped to this step.